

Madhav Suri

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EDUCATION

The University of Texas at Dallas

Richardson, TX

Bachelor of Science in Computer Science and Mathematics | GPA: 3.8/4.0

Expected Dec. 2027

- **Relevant Coursework:** Data Structures and Algorithms, OOP, Discrete Mathematics, Database Systems, Software Engineering, Operating Systems
- **Certifications:** AWS Cloud Practitioner, Google Project Management Certificate, CAPM in Progress

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, Dart, SQL, Bash, Go, HTML, JSON

Frameworks/Tools: React, Next.js, Node.js, FastAPI, REST APIs, Git, Firebase, PostgreSQL, Docker, Kubernetes, Jenkins, Linux/Unix, AWS, Tableau, Power BI

EXPERIENCE

Frontend Developer

Jan. 2026 – Present

ACM UTD

Richardson, TX

- Built the **React/TypeScript** frontend of an AI-powered academic-support platform across **5+ product workflows**, including office-hour scheduling, forum Q&A, self-check grading, and professor analytics, scaling to **350+ live users**.
- Integrated the **Anthropic Claude API** with prompt-engineering techniques for query triage and automated review, achieving 92% tier-1 automated resolution.
- Developed **45+ reusable UI components** and validated backend **REST APIs** with Postman while collaborating with ACM engineering teams and academic stakeholders.

Software & AI Engineering Intern

Aug. 2025 – Dec. 2025

SoftBank Robotics America

Southlake, TX

- Developed **EdgeSight**, a real-time assistive vision system running fully on a resource-constrained **NVIDIA Jetson Orin Nano** with 8GB unified memory, Linux/JetPack 6.2, and CUDA 12.6.
- Integrated **YOLOv8** OIV7 600-class detection, multi-object tracking, a multimodal VLM assistant, and a **FastAPI** dashboard, sustaining **15+ FPS** under CPU/GPU memory, power, and thermal limits.
- Reduced on-device VLM query latency by **~8x** from 74s to **~7–9s** by benchmarking 5 VLMs, including Gemma-4B, Qwen2.5-VL-3B, moondream, and Gemma-4 E4B/E2B, across **Ollama**, **vLLM**, and **llama.cpp**.
- Deployed a 4-bit-quantized **Gemma-4 E2B** model on **llama.cpp** with FlashAttention and q8_0 KV-cache, while accelerating detection through **TensorRT**, FP16 mixed precision, and MAXN_SUPER clock tuning.
- Profiled GPU-memory contention with **tegrastats**, identified vLLM's **~2.7GB** overhead as infeasible for the 8GB edge device, and designed a hybrid **YOLO + on-demand VLM + dynamic question-routing** pipeline.

Data Engineering Intern

May 2025 – Aug. 2025

Riddell

Irving, TX

- Engineered automated ETL pipelines in **Python** using pandas, NumPy, and SQL to process **5,000+** helmet-impact sensor readings per session into **AWS S3**, automating a previously manual data-preparation process.
- Built **scikit-learn**-powered telemetry reconciliation workflows against performance baselines, improving reporting and accelerating product-iteration evaluation cycles.

PROJECTS

RiskOS AI | *FastAPI, React, SQLAlchemy, Spark, DuckDB, scikit-learn, Pytest, S3*

[LINK](#)

- Architected a **fintech fraud-operations platform** using **FastAPI**, **React**, and **SQLAlchemy ORM** to process and route **6.3M transactions** into low-latency analyst queues.
- Built an **analytics layer** using **Parquet** datasets in **S3**, queried through **DuckDB** and **Spark**, to support experiment tracking and case-review workflows.
- Implemented **explainable risk scoring** with **scikit-learn Logistic Regression** and 7 deterministic validation rules, achieving **0.901 ROC-AUC**, 82.6% fraud recall, and 94.3% automated routing coverage.
- Added **RBAC**, manager-only overrides, **immutable audit trails**, and a **111-test Pytest** suite in a **GitHub Actions CI/CD** pipeline.

RepoPulse Agent | *Next.js, TypeScript, OpenAI API, GitHub API, Vitest, Docker*

[LINK](#)

- Engineered an **agentic AI codebase-intelligence platform** in **Next.js** and **TypeScript** with a **7-stage workflow** and human validation gates.
- Built **semantic code search** using **OpenAI embeddings**, syntax-tree parsing, **GitHub API** patch testing, Docker sandboxes, and **200+ Vitest checks**.
- Sustained **sub-1.5s latency** across **10,000+ file codebases** while cutting token spend by 35%.

D.I.Y.A. Academic Support Platform | *React, Vite, Node.js, Express, RAG, LangChain*

[LINK](#)

- Re-engineered an **AI full-stack academic platform** with group-scoped **RBAC** across student and professor routes, securing multi-tenant workflows.
- Built a course-material **Retrieval-Augmented Generation** system using MiniLM embeddings, recursive chunking, cosine search, **Pinecone**, **LangChain**, and OpenAI embeddings across **30+ documents**, achieving 95% retrieval relevance.

LEADERSHIP AND ACTIVITIES

Research Team Lead

Jan. 2026 – Present

The Erik Jonsson School of Engineering and Computer Science at UT Dallas

Richardson, TX

- Led undergraduate research under Prof. Yi Ding on **Trusted Execution Environments** and confidential ML inference, building secure C/C++/Python pipelines with **ARM TrustZone** and **Intel SGX**; presented at the Capital of Texas Undergraduate Research Conference; limited overhead to ~12% across 5 edge-AI workloads and merged **3 Open Enclave SDK PRs** (<https://doi.org/10.5281/zenodo.20840404>).